

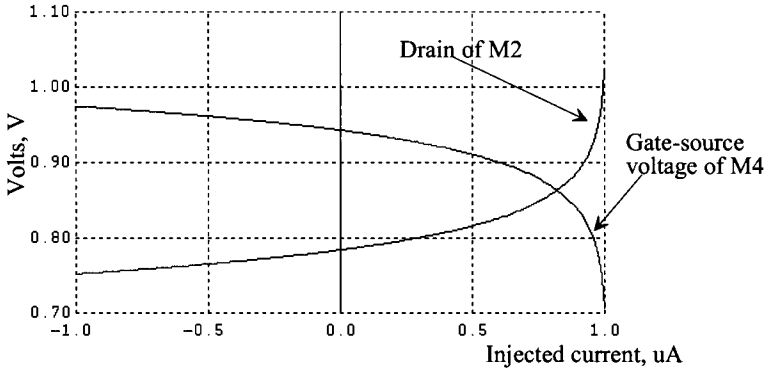


Figure 9.8 SPICE simulations verifying the discussions in Ex. 9.3.

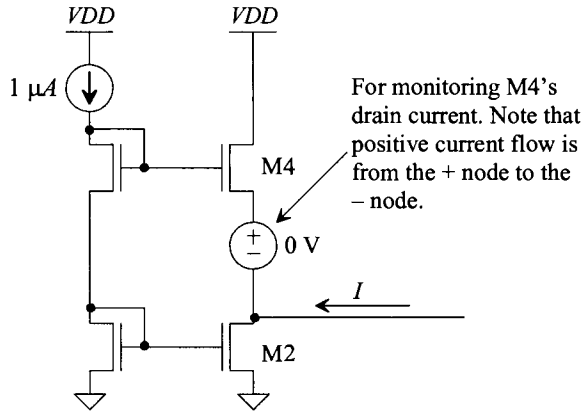


Figure 9.9 Adding zero volt sources to monitor currents in SPICE.

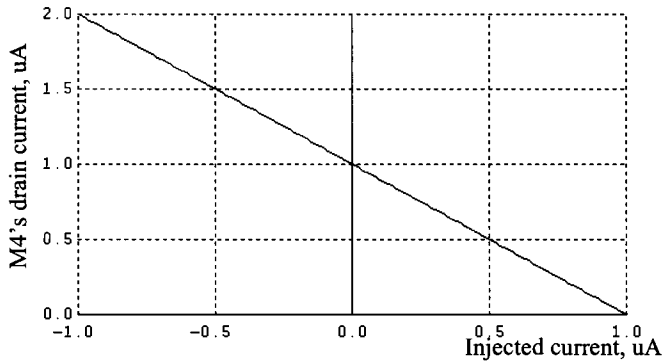


Figure 9.10 How the drain current of M4 changes with injected current.

Threshold Voltage and Body Effect

For an NMOS device, the threshold voltage increases when the source is at a higher potential than the NMOS body (the p-substrate, or ground, in this book). This change in threshold voltage is called the *body effect*. A simple example of a MOSFET operating with body effect (an increased threshold voltage) is seen in Fig. 9.7. Both M3 and M4 have a larger threshold voltage than M1 and M2.

From Ch. 6 the change in threshold voltage can be written as a function of the source to bulk potential, V_{SB} , Fig. 9.11, as

$$V_{THN}(V_{SB}) = V_{THN0} + \gamma_n \left(\sqrt{2|V_{fp}| + V_{SB}} - \sqrt{2|V_{fp}|} \right) \quad (9.10)$$

and for the PMOS (again we simply switch the subscripts so that all voltages and currents are positive)

$$V_{THP}(V_{BS}) = V_{THP0} + \gamma_p \left(\sqrt{2|V_{fn}| + V_{BS}} - \sqrt{2|V_{fn}|} \right) \quad (9.11)$$

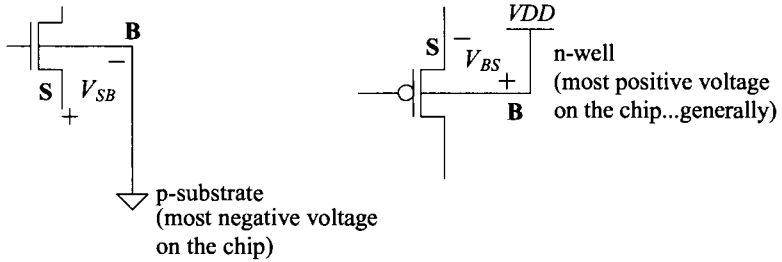


Figure 9.11 The body connection of a MOSFET.

Qualitative Discussion

It's useful to have a feeling for how the threshold voltage changes with increasing source-to-body potential. Figure 9.12 shows a plot of the threshold variation with V_{SB} . For large values of V_{SB} , the threshold voltage change isn't very significant, while for small values it is significant. The circuit seen in Fig. 9.13 is an example where we might want small variations in the threshold voltage. Because the resistor is large, the gate-source voltage will be close to the threshold voltage. If the threshold voltage didn't vary, then the drain current would be linearly related to the input voltage. This circuit is useful for voltage-to-current conversions.

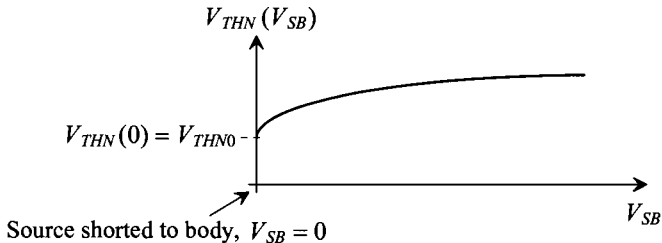


Figure 9.12 Variation in threshold voltage with source to bulk potential.

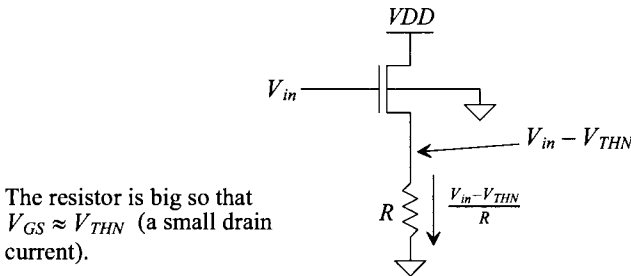


Figure 9.13 An example of a circuit where body effect is important.

Example 9.4

Simulate the operation of the voltage-to-current converter in Fig. 9.13 if the size of the MOSFET is 10/1 and the resistor is 10MEG. Use long-channel devices.

Figure 9.14a shows how the current varies through the MOSFET with V_{in} changing from 1 V to 5 V. We start at 1 V V_{in} because the current will shut off when $V_{in} < V_{THN}$, resulting in a large nonlinearity. Looking at the linearity of the current, it looks really good (straight). However, in Fig. 9.14b we take the derivative of the line in (a) to see the slope. The variation in the slope is approximately 20%. For precision design of voltage-to-current converters, the variation in the linearity of the threshold voltage can be a limiting factor. We should note that channel length modulation also contributes to nonlinearities (so increasing the length, to increase r_o , of the device can help improve the linearity).

Question: what is the ideal slope for Fig. 9.14; that is, what is the ideal value of dI/dV_{in} ? Answer: $1/R$ or, in this example, $100 \times 10^{-9} \text{ A/V}$. ■

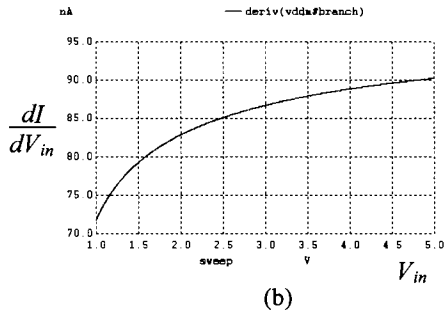
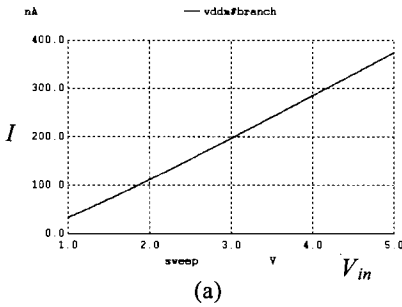


Figure 9.14 (a) Current flowing in the circuit of Fig. 9.13 and (b) its linearity.

There isn't any way to eliminate the body-effect in an n-well process where the NMOS device body is the substrate (though, if a triple well process is used, the NMOS devices can sit in their own wells). However, the PMOS devices can sit in their own wells and have separate bodies (and so we can design without body effect).

The Triode Region

The relationship between I_D , V_{GS} , and V_{DS} for an NMOS device operating in the triode region is

$$I_D = KP_n \frac{W}{L} \cdot \left((V_{GS} - V_{THN})V_{DS} - \frac{V_{DS}^2}{2} \right) \quad (9.12)$$

where $V_{GS} \geq V_{THN}$ and $V_{DS} \leq V_{DS,sat} (= V_{GS} - V_{THN})$. The equivalent expression for the PMOS device is

$$I_D = KP_p \frac{W}{L} \cdot \left((V_{SG} - V_{THP})V_{SD} - \frac{V_{SD}^2}{2} \right) \quad (9.13)$$

where $V_{SG} \geq V_{THP}$ and $V_{SD} \leq V_{SD,sat} (= V_{SG} - V_{THP})$.

We said earlier, in Fig. 9.4 and the associated discussion, that the MOSFET looks like a resistor when it is operating in the triode region. To estimate the value of the resistance, we can use

$$R_{ch}^{-1} = \frac{\partial I_D}{\partial V_{DS}} = KP_n \cdot \frac{W}{L} \cdot (V_{GS} - V_{THN}) - KP_n \cdot \frac{W}{L} \cdot V_{DS} \quad (9.14)$$

or

$$R_{ch} = \frac{1}{KP_n \cdot \frac{W}{L} \cdot (V_{DS,sat} - V_{DS})} \quad (9.15)$$

If $V_{DS,sat} \gg V_{DS}$, this equation can be written as

$$R_{ch} \approx \frac{1}{KP_n \cdot \frac{W}{L} (V_{GS} - V_{THN})} \quad (9.16)$$

The Cutoff and Subthreshold Regions

For general design, we normally assume that the device is off (meaning zero drain current) when $V_{GS} < V_{THN}$ or $V_{SG} < V_{THP}$. However, it would be more correct to say that the device is operating in the subthreshold region (instead of the strong inversion region we've discussed so far in this chapter). The subthreshold current of a MOSFET operating in the active region (the amplifying region, which is similar to the saturation region for a MOSFET operating in strong inversion) is modeled using

$$I_D = I_{D0} \cdot \frac{W}{L} \cdot e^{(V_{GS} - V_{THN})/nV_T} \quad (9.17)$$

$V_{GS} \leq V_{THN}$ and $V_{DS} > 4V_T$ (remembering the thermal voltage, V_T , is kT/q or 26 mV at room temperature). The current I_{D0} is the (scaled) current that flows when $V_{GS} - V_{THN} = 0$ (the gate-source voltage is equal to the threshold voltage). Note that Eq. (9.17) shows no dependence on V_{DS} . This indicates that the MOSFET has infinite output resistance when operating in the active region (which isn't the case, and so this equation has limitations).

9.1.2 Small Signal Models

A quick note concerning symbols: throughout the book we'll represent a signal containing both AC and DC components by a lowercase letter with uppercase subscripts. AC signals are represented with both lowercase letters and subscripts, while DC signals are represented with both uppercase letters and subscripts, see Fig. 9.15. (Note that if the maximum value of v_{GS} is V_{DD} in Fig. 9.15, then the MOSFET is either in the saturation region or off [$V_{GS} < V_{THN}$].)

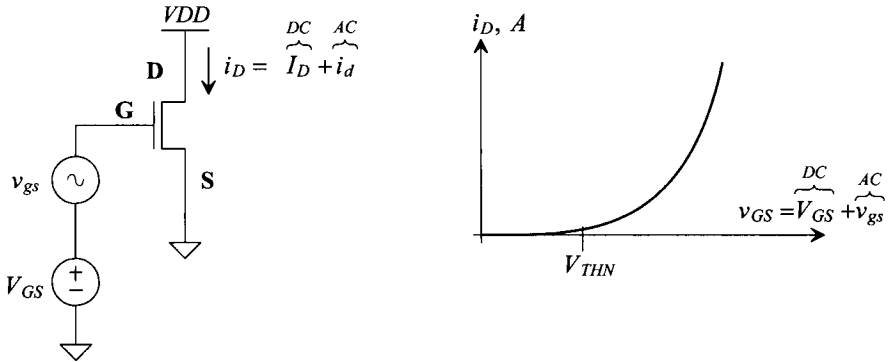


Figure 9.15 IV curves of a MOSFET in saturation.

Small-signal models are used to calculate AC gains. Figure 9.16 shows the basic idea. We adjust the DC gate-source voltage, V_{GS} , to a value that corresponds to a DC drain current I_D . At this bias point, we apply a small AC signal where $|v_{gs}| \ll V_{GS}$ and $|i_d| \ll I_D$. Because the signals are small, the change in drain current, i_d , with gate voltage, v_{gs} is essentially linear (as seen in the blown-up view in Fig. 9.16). If our AC signal amplitudes get comparable to the DC operating (or bias) points, we get high nonlinearity (which makes feedback necessary for any highly linear amplifier).

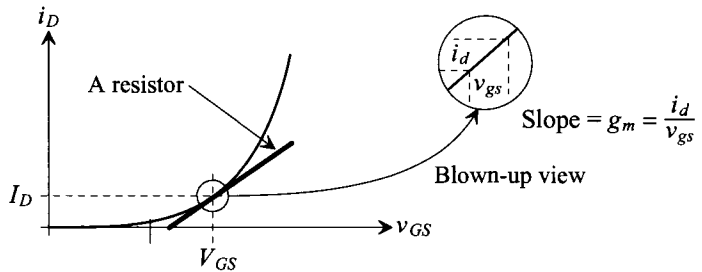


Figure 9.16 How small-signal parameters are calculated.

We often treat DC voltage sources as shorts when doing AC analysis (and current sources as opens). Looking at Fig. 9.15, we see that it's impossible for the AC signal to generate a voltage across the DC source. The voltage across the DC source is fixed with zero AC component so that the DC voltage source is an AC short.

Finally note that performing a small-signal analysis consists of the following steps:

- (1) Calculating the bias point of the circuit using the DC equations from Sec. 9.1.1.
- (2) Using the DC values from (1) to calculate the small-signal parameters. Small-signal AC parameters are always a function of the DC operating point.
- (3) Replacing the active elements (e.g., MOSFETs) with their small-signal models. At the same time, the DC sources are removed (that is, short out all DC voltage sources and open up all DC current sources).

An AC analysis doesn't include any DC voltages or currents. For example, suppose we have an (AC) $v_{gs} = 1$ mV. It *doesn't make sense* to say that the MOSFET is off because $v_{gs} < V_{THN}$. To perform a small-signal analysis (to calculate small-signal parameters), the MOSFETs are in saturation or triode (meaning $V_{GS} > V_{THN}$).

Transconductance

An extremely important parameter in analog design is a device's transconductance, g_m . The g_m of a device is an AC small-signal parameter that relates the AC gate voltage to the AC drain current, that is,

$$i_d = g_m \cdot v_{gs} \quad (9.18)$$

From Figs. 9.15 and 9.16, g_m is simply the slope of the line at the intersection of the DC operating values V_{GS} and I_D . Using Eq. (9.1), without concerning ourselves with channel-length modulation, we can write

$$i_D = i_d + I_D = \frac{KP_n}{2} \cdot \frac{W}{L} \cdot \left(\sqrt{v_{gs} + V_{GS} - V_{THN}} \right)^2 \quad (9.19)$$

To find the slope (g_m) of the i_D - v_{GS} curve at the fixed bias points V_{GS} and I_D (Fig. 9.16), we take the derivative of this equation with respect to the x-axis variable (v_{GS})

$$g_m = \left[\frac{\delta i_D}{\delta v_{GS}} \right]_{V_{GS} = \text{constant}}^{I_D = \text{constant}} = KP_n \cdot \frac{W}{L} \cdot (v_{gs} + V_{GS} - V_{THN}) \quad (9.20)$$

If we remember

$$\beta_n = KP_n \cdot \frac{W}{L} \quad \text{and} \quad |v_{gs}| \ll V_{GS} \quad (9.21)$$

then we can write

$$g_m = \beta_n \left(\sqrt{V_{GS} - V_{THN}} \right) = \sqrt{2\beta_n I_D} \quad (9.22)$$

The key points are that g_m goes up as the root of the drain current and linearly with $V_{DS,sat}$.

Example 9.5

Calculate the DC and AC voltages and currents for the circuit seen in Fig. 9.17. Use the long-channel MOSFET parameters from Ch. 6.

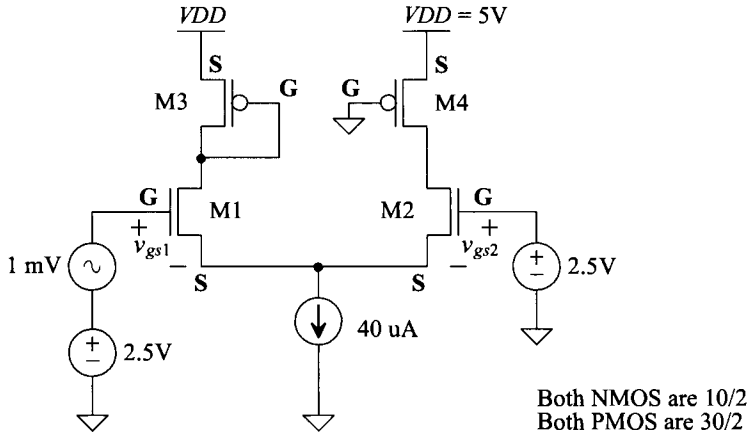


Figure 9.17 Circuit discussed in Ex. 9.5.

We begin by looking at the DC operating point. The gates of both M1 and M2 are at 2.5V. This is necessary to keep them turned on. Since the sources of M1 and M2 are physically tied together, $V_{GS1} = V_{GS2}$ and $I_{D1} = I_{D2} = 20 \mu A$. Rewriting Eq. (9.1), neglecting channel-length modulation, results in

$$V_{GS} = \sqrt{\frac{2I_D}{KP_n} \cdot \frac{L}{W}} + V_{THN} \quad (9.23)$$

Assuming both M1 and M2 are operating in the saturation region (we'll verify this in a moment), we get

$$V_{GS1} = V_{GS2} = \sqrt{\frac{40}{120} \cdot \frac{2}{10}} + 0.8 = 1.058 V \rightarrow V_{DS,sat} \approx 250 mV$$

The source-to-gate voltage for M3 (knowing its drain current is $20 \mu A$) is

$$V_{SG3} = \sqrt{\frac{2 \cdot 20}{40} \cdot \frac{2}{30}} + 0.9 = 1.158 V \rightarrow V_{SD,sat} \approx 250 mV$$

The drain potential of M1 and M3 is

$$V_{D1} = V_{D3} = V_{DD} - V_{SG3} = 3.842 V$$

From Fig. 9.5 and the associated discussion, we know that M3 is in saturation. To see if M1 is in saturation, we use Eq. (9.7)

$$V_{D1} \stackrel{?}{\geq} V_{G1} - V_{THN} \rightarrow 3.842 \geq 1.7 \text{ V (yes, M1 is in saturation)}$$

Next we look at M4. M4's source-to-gate voltage is 5 V. It's very likely that it is in triode. We know, for a PMOS to be operating in the saturation region, we must have

$$\underbrace{V_S - V_D}_{V_{SD}} \geq \underbrace{V_S - V_G}_{V_{SG}} - V_{THP} \rightarrow V_D \leq V_G + V_{THP} \quad (9.24)$$

M4's gate is grounded ($V_G = 0$), so for M4 to be in saturation, $V_D \leq 0.9 \text{ V}$. Since the gate of M2 is at 2.5 V and $V_{GS2} = 1.058$, then its source is 1.442 V, which makes it impossible for M4 to be saturated. To estimate the drain-to-source voltage of M4, we use Eq. (9.13), knowing that M4's drain current is 20 μA and its gate-source voltage is 5 V

$$20 = 40 \cdot \frac{30}{2} \cdot \left((5 - 0.9)V_{SD} - \frac{V_{SD}^2}{2} \right)$$

which results in $V_{SD} = 8.13 \text{ mV}$. The drains of M4 and M2 are then 4.992 V or essentially at V_{DD} . Clearly M2 is operating in the saturation region. M4 can be thought of as a resistor with a value of (Eq. [9.16] noting $V_{SD,sat} \gg V_{SD}$, that is, $4.1 \text{ V} \gg 8.13 \text{ mV}$)

$$R_{chM4} = \frac{1}{(40 \mu\text{A/V}) \cdot \frac{30}{2} \cdot 4.1} = 407 \Omega$$

Note that we could have estimated the resistance using $V_{SD}/I_D = 8.13\text{m}/20\mu = 407 \Omega$. Before moving on, let's do an operating point analysis for this circuit (an .op analysis). The SPICE netlist is seen below.

*** Example 9.5 CMOS: Circuit Design, Layout, and Simulation ***

```
.control
destroy all
run
** for the operating point analysis
print all
*
** for the AC analysis
*plot mag(vd13) mag(vs12) mag(vg1) mag(vd34)
*
** for the transient analysis
*plot vd13
*plot vs12
*plot vg1
.endc

.option scale=1u
.op
**.ac dec 100 1 10k
**.tran 1u 300u

VDD  VDD  0      DC    5
VG1   VG1  0      DC    2.5   AC    1m    SIN 2.5 1m 10k
VG2   VG2  0      Dc    2.5
lbias VS12  0      DC    40u
```

```

M1  VD13  VG1  VS12  0      NMOS L=2 W=10
M2  VD24  VG2  VS12  0      NMOS L=2 W=10
M3  VD13  VD13  VDD  VDD    PMOS L=2 W=30
M4  VD24  0     VDD  VDD    PMOS L=2 W=30

.MODEL NMOS NMOS LEVEL = 3
+ TOX  = 200E-10      NSUB = 1E17      GAMMA = 0.5
+ PHI  = 0.7          VTO  = 0.8        DELTA = 3.0
+ UO   = 650          ETA  = 3.0E-6     THETA = 0.1
+ KP   = 120E-6       VMAX = 1E5        KAPPA = 0.3
+ RSH  = 0            NFS  = 1E12       TPG  = 1
+ XJ   = 500E-9       LD   = 100E-9
+ CGDO = 200E-12      CGSO = 200E-12    CGBO = 1E-10
+ CJ   = 400E-6       PB   = 1          MJ   = 0.5
+ CJSW = 300E-12      MJSW = 0.5
*

.MODEL PMOS PMOS LEVEL = 3
+ TOX  = 200E-10      NSUB = 1E17      GAMMA = 0.6
+ PHI  = 0.7          VTO  = -0.9       DELTA = 0.1
+ UO   = 250          ETA  = 0           THETA = 0.1
+ KP   = 40E-6        VMAX = 5E4        KAPPA = 1
+ RSH  = 0            NFS  = 1E12       TPG  = -1
+ XJ   = 500E-9       LD   = 100E-9
+ CGDO = 200E-12      CGSO = 200E-12    CGBO = 1E-10
+ CJ   = 400E-6       PB   = 1          MJ   = 0.5
+ CJSW = 300E-12      MJSW = 0.5

.end

```

A portion of the operating point analysis output is

```

vd13 = 3.854977e+00 (we hand calculated 3.842)
vd24 = 4.989641e+00 (we hand calculated 4.992)
vs12 = 1.171189e+00 (we hand calculated 1.442)

```

The only significant difference between our hand calculations and the simulation results is the potential calculated for the sources of M1/M2. The smaller value in the simulation is because we didn't include the body effect in our calculations. We used $V_{THN0} = 0.8 V$ when the actual threshold voltage was closer to $1.1 V$.

Let's now turn our attention towards the AC analysis. The transconductance of M1 and M2 is

$$g_{m1} = g_{m2} = \sqrt{2 \cdot KP_n \frac{W}{L} \cdot I_D} = \sqrt{2 \cdot 120\mu \cdot \frac{10}{2} \cdot 20\mu} \approx 150 \mu A/V$$

The transconductance of M3 is

$$g_{m3} = \sqrt{2 \cdot KP_p \frac{W}{L} \cdot I_D} = \sqrt{2 \cdot 40\mu \cdot \frac{30}{2} \cdot 20\mu} \approx 150 \mu A/V$$

M4 is operating in the triode region and so we think of it as a resistor (407Ω). Figure 9.18 shows the simplified AC schematic of Fig. 9.17. Notice how we can replace M3 with a resistor of $1/g_{m3}$. This is because the AC voltage across M3 is $v_{sg3} = v_{sd3}$ and the AC current through it is i_d or

$$\frac{1}{g_m} = \frac{v_{sd}}{i_d} = \frac{v_{sg}}{i_d} \quad (9.25)$$

A gate-drain connected MOSFET with a current flowing through it is always in saturation, as discussed earlier, and can be thought of as a small-signal resistance of $1/g_m$ (again, remember this).

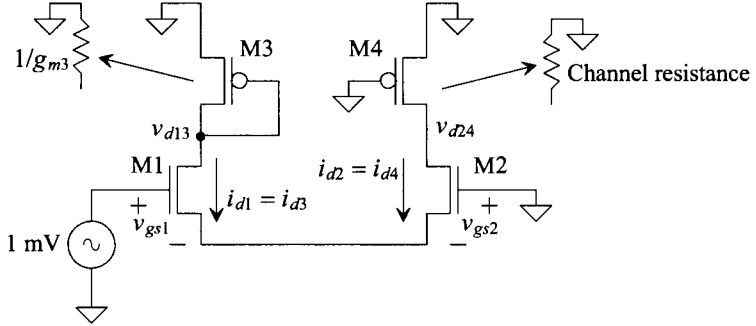


Figure 9.18 AC schematic for the circuit in Fig. 9.17.

When we talk about negative currents or voltages in an AC small-signal analysis, we are indicating that the overall AC + DC signal is decreasing. For example, we'll see in a moment that the AC small-signal drain current in M2, Fig. 9.18, is negative. This means that the current may be going from 20 μA (its DC operating point) to, say, 19.995 μA with an AC input. We would then say that the AC drain current is, i_{d2} , -5 nA.

To analyze the circuit, we can begin by writing

$$1 \text{ mV} = v_{gs1} - v_{gs2} = \frac{i_{d1}}{g_{m1}} - \frac{i_{d2}}{g_{m2}}$$

We know that $g_{m1} = g_{m2}$ and, from Fig. 9.18,

$$i_{d1} = -i_{d2}$$

so

$$v_{gs1} = -v_{gs2} = 0.5 \text{ mV}$$

The sources of M1 and M2 are at an AC voltage of 0.5 mV. The AC drain currents are

$$i_{d1} = i_{d3} = -i_{d2} = -i_{d4} = g_{m1} \cdot v_{gs1} = (150 \mu\text{A}/\text{V}) \cdot 0.5 \text{ mV} = 75 \text{ nA}$$

This means the overall (AC + DC) drain current of M1/M3 is

$$i_{D1} = 20 \mu\text{A} + 0.075 \sin 2\pi f t$$

where f is the frequency of our 1 mV input signal and

$$i_{D2} = 20 \mu\text{A} - 0.075 \sin 2\pi f t$$

noting that i_{D1} and i_{D2} must sum to the DC bias of 40 μA .

The (AC) drain voltage of M1 and M3 is

$$v_{d1} = v_{d3} = -i_{d1} \cdot \frac{1}{g_{m3}} = -\frac{75 \text{ nA}}{150 \mu\text{A/V}} = -0.5 \text{ mV}$$

The voltages on the drains of M2 and M4 are

$$v_{d2} = v_{d4} = -i_{d4} \cdot R_{chM4} = (75 \text{ nA}) \cdot 407 = 0.03 \text{ mV}$$

We'll verify these voltages with SPICE in a moment. This lengthy example illustrates several key concepts that will be used throughout the analysis and design of analog CMOS circuits in this book. Before moving on, make sure that the concepts are clear and well understood (SPICE can be very helpful for this).

Question: How would we look at the SPICE simulated currents (AC and DC) flowing in the MOSFETs in Fig. 9.17? Answer: add zero-volt voltage sources as seen in Fig. 9.9 and the associated discussion. ■

AC Analysis

We can perform a small-signal analysis using a SPICE “.ac” statement. Just like in our hand calculations, SPICE first calculates the operating point using the DC sources (no AC sources are present in the circuit). Then SPICE replaces the active devices with their small-signal equivalent circuits (no DC sources in the circuit, now just the AC sources). The key point here is that SPICE assumes the user knows that the AC components should be much smaller than the DC components. If we were to simulate, using an AC analysis, the operation of the circuit in Fig. 9.17, but used a 1,000 V AC input instead of a 1 mV AC input, SPICE would simply scale all of the outputs (the simulation won't tell the user that the AC input is too big).

The possible syntax for an .ac analysis statement is

```
.AC DEC ND FSTART FSTOP
.AC OCT NO FSTART FSTOP
.AC LIN NP FSTART FSTOP
```

where DEC stands for decade (our x-axis in an .ac analysis is frequency) and ND is the number of points per decade (OCT stands for octave and LIN stands for linear). FSTART and FSTOP are the starting and ending frequencies for the small-signal AC analysis.

We can do a SPICE AC analysis for the circuit in Fig. 9.17 by changing the .op analysis used in the netlist to an AC analysis control statement

```
.ac dec 100 1 10k
```

where we've picked a frequency range of 1 Hz to 10k Hz with 100 points calculated per decade. (The first decade is 1 to 10 Hz, the second, 10 Hz to 100 Hz, etc. If we had used a start frequency of 2 Hz, then the first decade would be 2 Hz to 20 Hz, the second would be 20 Hz to 200 Hz, noting that increasing by a decade is multiplying by 10 and decreasing by a decade is dividing by 10.) The input signal (the gate of M1) is

```
VG1 VG1 0 DC 2.5 AC 1m SIN 2.5 1m 10k
```